# Post-Bin $\ell^1$ Mass and Determinant-Energy Survival at a Fixed Shift: Sharp Two-Sided Exponent Transfer for the Matched Shell

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**Abstract**

Let $A_X(n) = A_{C_X}(n)$ be the actual signed normalized-determinant coefficient of the TPC-32 matched shell at one fixed nonzero shift $h_0$, with

$$C_X = \lfloor J_X \rfloor, \qquad N_{0,X} = J_X Q_X^2, \qquad Q_X = X^{267/400+o(1)}, \quad J_X = X^{133/400+o(1)}.$$

Write

$$S_X = \#\operatorname{supp} A_X, \quad L_X = \|A_X\|_\infty, \quad M_X = \|A_X\|_1, \quad D_X = \|A_X\|_2^2.$$

We prove the sharp finite-sequence inequalities

$$\frac{M_X^2}{S_X} \le D_X \le L_X M_X.$$

The verified physical support and singleton bounds of TPC-32 give

$$S_X \ll Q_X, \qquad L_X \ll X^{o(1)} \frac{N_{0,X}}{Q_X}.$$

It follows that

$$M_X \ge X^{-\mu+o(1)} N_{0,X} \quad \Longrightarrow \quad D_X \ge X^{-2\mu+o(1)} \frac{N_{0,X}^2}{Q_X},$$

whereas

$$D_X \ge X^{-\lambda+o(1)} \frac{N_{0,X}^2}{Q_X} \quad \Longrightarrow \quad M_X \ge X^{-\lambda+o(1)} N_{0,X}.$$

Thus subpower survival of the natural determinant energy is equivalent to subpower survival of the actual post-bin $\ell^1$ mass. The same energy hypothesis forces

$$S_X \ge X^{-\lambda-o(1)} Q_X.$$

The mass-to-energy exponent is sharp, and the reverse mass and support exponents are sharp in the nontrivial range $X^{-\lambda} Q_X \to \infty$, which contains the physical endpoint range $\lambda < 1/400$.

The word post-bin is essential. A raw atomic mass, a pre-bin absolute mass, or the absolute matched majorant may upper-bound $M_X$, but none can lower-bound it: opposite atoms in one determinant fiber can cancel completely. The result therefore closes an exact L1 equivalence between two literal statistics; it does not prove that either statistic survives at its natural scale, and it gives no zero-frequency cancellation estimate. In particular, it proves no L2 arithmetic input, parity result, prime-pair lower bound, or twin-prime conclusion.

**Keywords:** fixed shift; determinant coefficient; post-bin mass; energy survival; sharp norm interpolation; matched shell; endpoint audit.

**Convention.** Every arithmetic statement concerns the high-$\beta$ TPC-32 post-bin coefficient at one fixed nonzero $h_0$, with $Q_X = X^{267/400+o(1)}$, $J_X = X^{133/400+o(1)}$, and $C_X = \lfloor J_X \rfloor$. Thus $C_X \le J_X \ll Q_X$, as required by the imported determinant envelope. Complete native aggregation, exact active support, and the original global normalization are retained. The post-bin $\ell^1$ mass is never replaced by a pre-bin or raw absolute majorant.

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## 1 Introduction

The TPC-81 transfer architecture isolates the determinant energy

$$D_X = \sum_n |A_{C_X}(n)|^2$$

as one of the two remaining arithmetic statistics in the matched fixed-shift route [3]. TPC-82 then records the zero-slack relation between energy survival and distinguished zero-mode cancellation [4]. A lower bound for $D_X$ is not presently known at the natural scale. Before returning to the more elaborate pre-bin completion machinery, it is useful to ask a strictly post-bin question:

> How much of the determinant energy problem is already equivalent to survival of the actual post-bin $\ell^1$ mass?

For the literal coefficient, the answer is exact and asymmetric. A lower $\ell^1$ mass with loss $\mu$ yields an energy lower bound with loss $2\mu$. Conversely, an energy lower bound with loss $\lambda$ yields an $\ell^1$ lower bound with the same loss $\lambda$. At subpower resolution, the two survival statements are equivalent. This uses only two physical upper bounds already proved in TPC-32: the determinant support has size $O(Q_X)$, and a single determinant bin has amplitude at most $X^{o(1)}N_{0,X}/Q_X$ [2].

The result is elementary, but it closes a useful interface. It replaces the broad question "does the energy survive binning?" by the equivalent subpower question "does a natural amount of $\ell^1$ mass survive after binning?" This is not a replacement of arithmetic by norm interpolation. TPC-32 supplies only the upper bound

$$M_X \ll X^{o(1)}N_{0,X};$$

the matching lower bound remains open. Moreover, $M_X$ is formed after all literal atoms with the same determinant have been summed. A large raw absolute mass gives no such lower bound.

The paper has five main conclusions.

(i) We prove the sharp universal inequalities

$$M^2/S \le D \le LM$$

for a finite complex sequence, including all equality cases.

(ii) We insert the actual TPC-32 support and singleton bounds at $C_X = \lfloor J_X \rfloor$, obtaining the two-sided exponent transfer.

(iii) We prove that natural determinant energy forces a quantitatively large actual support. Thus energy cannot survive at its natural scale on a subpower-small fraction of the available determinant range.

(iv) We give extremizing model sequences showing that the factors $2\mu$, $\lambda$, and the support exponent cannot be improved from the stated data.

(v) We separate the actual post-bin mass from every pre-bin or raw absolute majorant and give exact cancellation counterexamples.

The distinction from the distinguished zero frequency is equally important. The mass

$$M_X = \sum_n |A_X(n)|$$

forgets the phases between different determinant bins. The zero mode

$$Z_X = \sum_n A_X(n)$$

depends on those phases. Two sequences can have identical $(S, L, M, D)$ and opposite zero-mode behavior. Therefore the present equivalence advances the energy side only; it neither proves nor models zero-frequency cancellation.

All physical statements keep one prescribed $h_0 \neq 0$, the actual coefficient $A_{C_X}$, the complete matched aggregation, and the global TPC-32 normalization. No assertion is specialized to $h_0 = 2$.

## 2 The literal matched determinant coefficient

Fix once and for all a nonzero integer shift $h_0$. We work on the high-$\beta$ TPC-32 matched packet

$$Q_X = X^{267/400+o(1)}, \qquad J_X = X^{133/400+o(1)}. \tag{1}$$

In particular $J_X \ll Q_X$. Put

$$C_X = \lfloor J_X \rfloor, \qquad N_{0,X} = J_X Q_X^2. \tag{2}$$

Hence $C_X \le J_X$ and $C_X \ll Q_X$, so the imported TPC-32 support, singleton, and energy estimates apply in their stated range. All fixed support and smoothness data are suppressed from the notation; implied constants may depend on them and on $h_0$, but not on $X$.

The signed normalized-determinant coefficient is

$$A_X(n) := A_{C_X}(n), \qquad n \in \mathbb{Z}, \tag{3}$$

where $A_{C_X}$ is formed only after the three matched physical channels and every native contribution in the determinant fiber have been summed. Define

$$\begin{aligned} S_X &= \#\operatorname{supp} A_X, \\ L_X &= \|A_X\|_\infty, \\ M_X &= \|A_X\|_1 = \sum_n |A_X(n)|, \\ D_X &= \|A_X\|_2^2 = \sum_n |A_X(n)|^2. \end{aligned} \tag{4}$$

When $A_X = 0$, all four quantities are zero. Every division by $S_X$ below is restricted to the nonzero case.

**Proposition 2.1** (Verified TPC-32 envelope)**.** *For the literal coefficient in* (3)*,*

$$S_X \ll Q_X, \tag{5}$$

$$M_X \ll X^{o(1)} N_{0,X}, \tag{6}$$

$$L_X \ll X^{o(1)} \frac{N_{0,X}}{Q_X}, \tag{7}$$

$$D_X \ll X^{o(1)} \frac{N_{0,X}^2}{Q_X}. \tag{8}$$

*Proof.* TPC-32 proves that the determinant support is contained in a fixed multiple of $[-Q_X, Q_X] \cap (\mathbb{Z} \setminus \{0\})$, giving (5). It also constructs a post-fiber absolute majorant $A^+_{C_X}$ with

$$\sum_n A^+_{C_X}(n) \ll X^{o(1)} N_{0,X}$$

and

$$\max_n A^+_{C_X}(n) \ll X^{o(1)} Q_X (C_X + J_X).$$

Since $C_X \le J_X$ and $N_{0,X}/Q_X = J_X Q_X$, the second estimate is

$$\max_n A^+_{C_X}(n) \ll X^{o(1)} \frac{N_{0,X}}{Q_X}.$$

The literal inequalities

$$|A_X(n)| \le A^+_{C_X}(n)$$

give (6) and (7). Finally,

$$D_X \le L_X M_X$$

gives (8); this is also the energy estimate stated in TPC-32 [2]. $\square$

*Remark* 2.2 (The atom bound is already normalized). The bound (7) is used only in the normalization in which TPC-32 proves it for the complete matched coefficient. It is not divided by a number of channels, triplets, fibers, or components, and no diagonal or componentwise normalization is inserted. Any alternative normalization would require a new literal comparison theorem.

*Remark* 2.3 (Actual versus available support). The number $S_X$ is the exact nonzero support after determinant aggregation. The interval of length $O(Q_X)$ supplies only the upper bound (5). It does not say that $S_X \asymp Q_X$; a lower support bound will instead follow conditionally from energy survival in corollary 4.4.

# 3 A sharp finite-sequence theorem

The algebra is independent of arithmetic and is stated over $\mathbb{C}$.

**Theorem 3.1** (Mass–energy sandwich). *Let $a = (a_n)$ be a nonzero finitely supported complex sequence and write*

$$S = \#\operatorname{supp} a, \qquad L = \|a\|_\infty, \qquad M = \|a\|_1, \qquad D = \|a\|_2^2.$$

*Then*

$$\boxed{\frac{M^2}{S} \le D \le LM.} \tag{9}$$

*Equivalently,*

$$M \ge \frac{D}{L}, \qquad S \ge \frac{D}{L^2}. \tag{10}$$

*Equality holds in either inequality of (9) if and only if $|a_n|$ is constant on $\operatorname{supp} a$. In that case both inequalities are equalities.*

*Proof.* Cauchy–Schwarz on the exact support gives

$$M^2 = \left(\sum_{n\in\operatorname{supp} a} |a_n|\right)^2 \le S \sum_{n\in\operatorname{supp} a} |a_n|^2 = SD.$$

This is the first inequality. For the second,

$$D = \sum_n |a_n|^2 \le L \sum_n |a_n| = LM.$$

The first formula in (10) is a rearrangement of $D \le LM$. Also

$$D = \sum_{n\in\operatorname{supp} a} |a_n|^2 \le SL^2,$$

which gives the second formula.

Equality in Cauchy–Schwarz occurs exactly when the vector $(|a_n|)_{n\in\operatorname{supp} a}$ is proportional to the all-ones vector. Equality in $D \le LM$ occurs exactly when each nonzero $|a_n|$ equals $L$. These are the same constant-modulus condition. This is the classical equality case of the elementary norm inequalities [1]. □

*Remark* 3.2 (Phases are unrestricted). The theorem uses only the moduli of the post-bin entries. The phases of the nonzero $a_n$ can be arbitrary. Consequently equality in (9) says nothing about the zero mode $\sum_n a_n$.

**Corollary 3.3** (Effective support). *The quantity*

$$S_{\mathrm{eff}}(a) := \frac{M^2}{D}$$

*satisfies*

$$1 \le S_{\mathrm{eff}}(a) \le S.$$

*It equals $S$ precisely for constant-modulus support.*

*Proof.* The upper bound is the first inequality of (9). The lower bound follows from $M^2 \geq D$, since

$$\left(\sum_n |a_n|\right)^2 \geq \sum_n |a_n|^2.$$

The equality characterization follows from theorem 3.1. ☐

## 4 Sharp exponent transfer on the matched shell

We now apply theorem 3.1 to the literal coefficient $A_X$. Every occurrence of $o(1)$ may denote a different function tending to zero.

**Theorem 4.1** (Two-sided post-bin exponent transfer)**.** *Let $A_X = A_{C_X}$ be the fixed-$h_0$ matched coefficient of section 2, and assume the verified envelope proposition 2.1.*

*(i) If, for some fixed $\mu \geq 0$,*

$$M_X \geq X^{-\mu+o(1)} N_{0,X}, \tag{11}$$

*then*

$$\boxed{D_X \geq X^{-2\mu+o(1)} \frac{N_{0,X}^2}{Q_X}.} \tag{12}$$

*(ii) If, for some fixed $\lambda \geq 0$,*

$$D_X \geq X^{-\lambda+o(1)} \frac{N_{0,X}^2}{Q_X}, \tag{13}$$

*then*

$$\boxed{M_X \geq X^{-\lambda+o(1)} N_{0,X}.} \tag{14}$$

*(iii) Under (13), the exact determinant support satisfies*

$$\boxed{S_X \geq X^{-\lambda-o(1)} Q_X.} \tag{15}$$

*Proof.* By (5) and the first half of (9),

$$D_X \geq \frac{M_X^2}{S_X} \gg X^{-2\mu+o(1)} \frac{N_{0,X}^2}{Q_X},$$

which proves (i), after absorbing fixed constants into $X^{o(1)}$.

For (ii), use $D_X \leq L_X M_X$ and (7):

$$M_X \geq \frac{D_X}{L_X} \gg \frac{X^{-\lambda+o(1)} N_{0,X}^2/Q_X}{X^{o(1)} N_{0,X}/Q_X} = X^{-\lambda+o(1)} N_{0,X}.$$

Finally, $D_X \leq S_X L_X^2$. Hence

$$S_X \geq \frac{D_X}{L_X^2} \gg \frac{X^{-\lambda+o(1)} N_{0,X}^2/Q_X}{X^{o(1)} N_{0,X}^2/Q_X^2} = X^{-\lambda-o(1)} Q_X.$$

This proves (iii). ☐

*Definition* 4.2 (Subpower natural survival)*.* We say that the post-bin mass survives at its natural scale if

$$M_X \geq X^{-o(1)} N_{0,X}. \tag{16}$$

We say that the determinant energy survives at its natural scale if

$$D_X \geq X^{-o(1)} \frac{N_{0,X}^2}{Q_X}. \tag{17}$$

These are lower-bound statements. Their matching upper bounds are already contained in proposition 2.1.

**Corollary 4.3** (Subpower equivalence)**.** *For the literal matched coefficient,*

$$\boxed{M_X \geq X^{-o(1)} N_{0,X} \quad \Longleftrightarrow \quad D_X \geq X^{-o(1)} \frac{N_{0,X}^2}{Q_X}.} \tag{18}$$

*Proof.* If $M_X \geq X^{-o(1)} N_{0,X}$, then

$$D_X \geq \frac{M_X^2}{S_X} \geq X^{-o(1)} \frac{N_{0,X}^2}{Q_X}$$

by (5). Conversely, if the displayed determinant-energy lower bound holds, then

$$M_X \geq \frac{D_X}{L_X} \geq X^{-o(1)} N_{0,X}$$

by (7). Thus no varying "fixed exponent" is being inserted into theorem 4.1. □

**Corollary 4.4** (Natural energy forces broad actual support)**.** *If* (17) *holds, then*

$$S_X \geq X^{-o(1)} Q_X.$$

*Together with $S_X \ll Q_X$, the actual support then occupies the available determinant scale up to subpower loss.*

*Proof.* Repeat the proof of part (iii) of theorem 4.1 with the subpower lower bound (17). □

*Remark* 4.5 (The asymmetry is real). A mass loss $\mu$ may be spread evenly across $Q_X$ bins, so the energy loss doubles to $2\mu$. In the reverse direction, energy can be carried by only $X^{-\lambda} Q_X$ bins at the maximal allowed amplitude, giving mass loss $\lambda$. The model sequences in section 5 make both mechanisms exact.

# 5 Sharpness at every exponent

The constant-modulus equality cases of theorem 3.1 already show sharpness in finite dimension. We now match the arithmetic normalization and show that the exponent transfers cannot be improved from only

$$S_X \ll Q_X, \qquad L_X \ll X^{o(1)} N_{0,X}/Q_X.$$

Rounding an integer support size changes only fixed or subpower factors and will be suppressed.

**Proposition 5.1** (Sharpness of mass-to-energy loss)**.** *Fix $\mu \geq 0$. There are finite sequences with*

$$S_X \asymp Q_X, \qquad L_X \leq \frac{N_{0,X}}{Q_X},$$

*for which*

$$M_X = X^{-\mu} N_{0,X}, \qquad D_X \asymp X^{-2\mu} \frac{N_{0,X}^2}{Q_X}.$$

*Thus the exponent $2\mu$ in* (12) *is best possible.*

*Proof.* Choose $S_X = \lceil Q_X \rceil$ nonzero coordinates and give each coordinate modulus

$$t_X = \frac{X^{-\mu} N_{0,X}}{S_X}.$$

Then

$$M_X = S_X t_X = X^{-\mu} N_{0,X}$$

and

$$D_X = S_X t_X^2 = \frac{X^{-2\mu} N_{0,X}^2}{S_X} \asymp X^{-2\mu} \frac{N_{0,X}^2}{Q_X}.$$

Also $t_X \ll N_{0,X}/Q_X$. Arbitrary unit phases may be attached to the coordinates. □

**Proposition 5.2** (Sharpness of energy-to-mass and support losses)**.** *Fix $\lambda \ge 0$ in a range for which $X^{-\lambda} Q_X \to \infty$. There are finite sequences satisfying*

$$L_X = \frac{N_{0,X}}{Q_X}$$

*and*

$$S_X \asymp X^{-\lambda} Q_X, \tag{19}$$

$$M_X \asymp X^{-\lambda} N_{0,X}, \tag{20}$$

$$D_X \asymp X^{-\lambda} \frac{N_{0,X}^2}{Q_X}. \tag{21}$$

*Consequently, neither the mass exponent $\lambda$ nor the support exponent $\lambda$ in theorem 4.1 can be improved from the stated envelope.*

*Proof.* Choose

$$S_X = \lfloor X^{-\lambda} Q_X \rfloor$$

coordinates, all of modulus $N_{0,X}/Q_X$. Then (19)–(21) follow by direct calculation. This sequence has constant modulus on its support and therefore attains equality on both sides of (9). □

*Remark* 5.3 (Very sparse range). If $X^{-\lambda} Q_X$ is bounded, the lower support conclusion is at most a constant-scale statement. The theorem remains valid, but there is no stronger power-scale support conclusion to extract from the envelope alone.

**Example 5.4** (Why the support upper bound is needed)**.** *Let $M = 1$ and spread this mass uniformly over $K$ coordinates. Then $D = 1/K$. Allowing $K$ to exceed the physical $O(Q_X)$ determinant range would make energy arbitrarily smaller than the scale in (12). Thus the first transfer uses the verified physical support interval, not merely finiteness.*

**Example 5.5** (Why the singleton bound is needed)**.** *A one-point sequence of amplitude $T$ has*

$$D = T^2, \qquad M = T, \qquad L = T.$$

*Without an upper bound for $L$ in the physical normalization, an energy lower bound does not determine the normalized mass scale $N_{0,X}$. The reverse transfer therefore genuinely uses the TPC-32 singleton estimate.*

# 6 Why raw and pre-bin masses cannot be substituted

The distinction between a post-bin coefficient and an atomic majorant is exact. Let $\mathcal{Y}_X$ be a native pre-bin key set, let $\nu_X : \mathcal{Y}_X \to \mathbb{Z}$ be the determinant label, and write the literal atomic contribution as $b_X(y)$. In the simplest unit-multiplier model,

$$A_X(n) = \sum_{\substack{y \in \mathcal{Y}_X \\ \nu_X(y) = n}} b_X(y). \tag{22}$$

The same discussion applies with archived multipliers absorbed into $b_X(y)$.

Define the fiberwise absolute majorant and raw absolute mass by

$$A_X^+(n) = \sum_{\substack{y \in \mathcal{Y}_X \\ \nu_X(y) = n}} |b_X(y)|, \qquad M_{\mathrm{raw},X} = \sum_{y \in \mathcal{Y}_X} |b_X(y)|. \tag{23}$$

Then

$$M_X = \sum_n |A_X(n)| \le \sum_n A_X^+(n) = M_{\mathrm{raw},X}. \tag{24}$$

**Proposition 6.1** (No reverse raw-mass inequality)**.** *There is no positive universal constant $c$ such that*

$$M_X \geq c\, M_{\mathrm{raw},X}$$

*for all literal determinant aggregations, even when the raw support, raw absolute values, determinant fiber, and global normalization are fixed.*

*Proof.* Put two atoms in the same determinant fiber and take their contributions to be 1 and $-1$. Then

$$M_{\mathrm{raw},X} = 2, \qquad A_X(n) = 0, \qquad M_X = D_X = 0.$$

Scaling both atoms makes the raw mass arbitrarily large while the post-bin coefficient remains zero. $\square$

**Corollary 6.2** (Majorants give upper bounds only)**.** *The two TPC-32 majorant statistics*

$$\sum_n A^+_{C_X}(n) \quad \textit{and} \quad \max_n A^+_{C_X}(n)$$

*validly imply upper bounds for $M_X$ and $L_X$. They do not imply a lower bound for $M_X$, $D_X$, or $S_X$.*

*Proof.* The upper implications follow from $|A_X(n)| \leq A^+_{C_X}(n)$. The counterexample in proposition 6.1 rules out every positive reverse implication. $\square$

*Remark* 6.3 (No channelwise renormalization)*.* The cancellation in proposition 6.1 occurs before the post-bin statistics are measured. Renormalizing each atom, channel, triplet, or component separately would erase that cancellation and change the coefficient. The only allowed $M_X$ is the norm of the already aggregated $A_X$ in the original global normalization.

## 6.1 Mass and energy do not control the zero mode

Let

$$Z_X = \sum_n A_X(n)$$

be the distinguished zero-frequency coefficient.

**Proposition 6.4** (Identical norm data, different zero modes)**.** *For every even $S$ and every $L > 0$, there are two sequences with identical $(S, L, M, D)$ but zero modes of magnitudes $SL$ and $0$, respectively.*

*Proof.* On a support of size $S$, take

$$a_n = L$$

for every coordinate and take

$$b_n = (-1)^n L.$$

Both sequences satisfy

$$\#\,\mathrm{supp} = S, \qquad \|\cdot\|_\infty = L, \qquad \|\cdot\|_1 = SL, \qquad \|\cdot\|_2^2 = SL^2.$$

The first zero mode is $SL$, while the second is zero. $\square$

*Remark* 6.5 (No flatness theorem)*.* Proposition 6.4 shows that even exact knowledge of all four post-bin statistics in (4) does not prove the TPC-32 distinguished-frequency flatness condition. The current paper concerns energy survival, not cancellation among determinant bins.

## 6.2 Observation versus premise

Numerically observing $M_X$ or $D_X$ on a finite cutoff is useful for locating possible loss. It is not a growing-$X$ lower bound. Likewise, an exact upper bound for the raw majorant is not evidence that the post-bin mass is close to that upper bound. A valid arithmetic input must be stated directly for $M_X$ or $D_X$ with the fixed $h_0$, the literal coefficient, and the original quantifiers.

## 7 The exponent and endpoint audit

### 7.1 What exponent enters the energy ledger

Suppose a future argument proves

$$M_X \geq X^{-\mu+o(1)} N_{0,X}.$$

Theorem 4.1 certifies the determinant-energy loss

$$\lambda_D \leq 2\mu \tag{25}$$

in the notation

$$D_X \geq X^{-\lambda_D+o(1)} \frac{N_{0,X}^2}{Q_X}.$$

This is a one-way certified upper bound on the loss exponent; the actual energy may be larger. Conversely, an independently proved energy loss $\lambda_D$ certifies a post-bin mass loss at most $\lambda_D$.

The TPC-82 zero-slack ledger compares this energy loss with a distinguished-zero-mode gain $\eta_Z$:

$$2\eta_Z \geq \lambda_D. \tag{26}$$

Thus, if the mass route is the only energy input and there are no additional energy losses, the sufficient compatibility condition is

$$\eta_Z \geq \mu.$$

If an auxiliary squared-energy loss $\lambda_{\mathrm{aux}}$ is present, the recorded condition becomes

$$2\eta_Z \geq 2\mu + \lambda_{\mathrm{aux}}.$$

The present paper proves none of these lower or cancellation hypotheses; it only identifies their exact transfer.

### 7.2 The strict physical loss rule

The $1/400$ appearing at the high-row TPC-32 flatness endpoint is not an allowance that may be reused for mass, energy, dictionary, or cover losses. In the full physical comparison, every fixed-power loss must be derived in the same literal carrier and counted once. Writing their complete sum as $\Lambda_{\mathrm{phys}}$, the operational rule remains

$$\boxed{\Lambda_{\mathrm{phys}} < \frac{1}{400} \quad \text{to continue;} \qquad \Lambda_{\mathrm{phys}} \geq \frac{1}{400} \quad \text{to stop.}} \tag{27}$$

Equality is in the stopping case. An open term cannot be assigned zero loss, and (25) cannot be counted twice under both the mass and energy names.

### 7.3 The L0/L1/L2 ledger

*Definition* 7.1 (Stage labels). (1) L0 is finite algebra independent of the physical carrier.

(2) L1 is an exact interface retaining the fixed nonzero $h_0$, literal coefficient, complete native aggregation, actual support, global normalization, and original quantifiers.

(3) L2 is a new asymptotic arithmetic estimate for that literal growing carrier, strong enough to reduce a positive exponent in the physical ledger.

**Proposition 7.2** (Stage classification of this paper)**.** *Theorem 3.1 is an exact* L0 *theorem. Theorem 4.1 and corollary 4.3 form an exact* L1 *equivalence for the actual TPC-32 matched coefficient. This paper proves no unconditional* L2 *estimate.*

*Proof.* The finite theorem uses no arithmetic. The transfer inserts the verified literal support and singleton bounds while preserving every item in definition 7.1(2). However, neither (11) nor (13) is proved here for any growing physical packet. Hence no new asymptotic lower bound is available to enter the L2 ledger. □

*Remark* 7.3 (What would count as the next advance). A proof of

$$M_X \geq X^{-o(1)} N_{0,X}$$

for the actual fixed-$h_0$ coefficient would, by corollary 4.3, close the natural determinant-energy gate. A lower bound for a raw majorant, an averaged shift, a selected subsequence of cutoffs, or a renormalized component would not.

# 8 Final claim boundary

**Theorem 8.1** (Post-bin mass–energy equivalence). *For the actual TPC-32 coefficient*

$$A_X = A_{\lfloor J_X \rfloor}$$

*at one fixed nonzero $h_0$, with $N_{0,X} = J_X Q_X^2$, the following are equivalent:*

$$M_X \geq X^{-o(1)} N_{0,X}, \tag{28}$$

$$D_X \geq X^{-o(1)} \frac{N_{0,X}^2}{Q_X}. \tag{29}$$

*More generally, a mass loss $\mu$ certifies energy loss $2\mu$, and an energy loss $\lambda$ certifies mass loss $\lambda$. The energy hypothesis also forces*

$$S_X \geq X^{-\lambda-o(1)} Q_X.$$

*The mass-to-energy exponent is sharp under the verified support envelope. The reverse mass and support exponents are sharp whenever $X^{-\lambda} Q_X \to \infty$, in particular throughout the relevant endpoint range $\lambda < 1/400$.*

*Proof.* This is theorem 4.1 and corollary 4.3 together with the extremizers of propositions 5.1 and 5.2. □

The theorem reduces one literal energy question to one literal post-bin mass question. It does not estimate either quantity from below. In particular, the TPC-32 absolute majorant and every pre-bin/raw absolute mass remain upper data only.

**Explicit exclusions.** This paper proves no lower bound for $M_X$ or $D_X$, no estimate for the distinguished zero mode $Z_X$, no fixed-shift Möbius cancellation theorem, and no result specialized to $h_0 = 2$. It does not prove a breach of sieve parity, a Hardy–Littlewood asymptotic, a prime-pair lower bound, or the twin-prime conjecture. Its complete contribution is the sharp L1 equivalence and its stop rules.

**Next gate.** The next positive task is to prove the following lower bound directly for the literal post-bin mass:

$$M_X \geq X^{-o(1)} N_{0,X}.$$

The alternative is an exact obstruction showing that the physical fiber geometry permits a fixed-power or complete post-bin mass loss. Either outcome must be stated for the same $h_0$, actual support, and global normalization. The full-route loss remains subject to (27).

## References


[1] G. H. Hardy, J. E. Littlewood, and G. Pólya. *Inequalities*. Cambridge University Press, second edition, 1952.

[2] Liang Wang. Matched prime–Möbius cutoff shells: Rank–two factorization, determinant–frequency dispersion, and the distinguished zero–frequency gate, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-32-matched-cutoff-frequency-gate. Preprint.

[3] Liang Wang. A fixed-shift arithmetic transfer gap: From literal completion certificates to zero-frequency flatness, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-81-fixed-shift-arithmetic-transfer-gap. Preprint.

[4] Liang Wang. A ten-paper fixed-shift completion-to-flatness audit: Exact certificate closures, the determinant-energy gate, and a zero-slack endpoint ledger, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-82-ten-paper-completion-to-flatness-audit. Preprint.